UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

DECEMBER EXAMINATION

UCCD2103 OPERATING SYSTEMS

THURSDAY, 22 DECEMBER 2022 TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

SECTION A (Answer All Questions)

Q1. (a) Draw and label the diagram of 7-State process model. (7 marks)

(b) Referring to the following scenarios, list each state transition of Process A based on the 7-state model:

(i) Process A is created and executes. (1 mark)

(ii) Then, Process A reads File Z from the disk storage. (1 mark)

(iii) Next, Process A is swapped out to the secondary memory due to insufficient main memory. (1 mark)

(iv) The main memory is available and Process A is swapped in to the main memory. (1 mark)

(v) File Z is ready. (1 mark)

(vi) Finally, Process A resumes execution until finish. (1 mark)

(c) Define the following terms:

(i) Mutual exclusion (2 marks)

(ii) Starvation (2 marks)

(iii) Race condition (2 marks)

(d) For each of the following, determine the type of real-time task (hard or soft real-time) in the application or system and briefly justify your answer:

(i) The main task of a smart watch is to track the wearer's heart rate and the number of footsteps the wearer has taken. (3 marks)

(ii) The main task of a vehicle's airbag system is to detect the intensity of the impact from impact sensors reading. Calculation is then made to determine if there is a need to deploy the airbag. (3 marks)

[Total : 25 marks]

Q2. (a) This question is about the Completely Fair Scheduler (CFS) found in Linux kernel. Assume that the system has a uni-processor with a red-black tree. There are two CPU-bound processes in Ready state; A and B with the following parameters shown in Table 1:

Table 1: Virtual runtimes and nice values of Process A and B

Parameters	Process A	Process B
Virtual runtime (nanoseconds)	1,283,888	1,283,642
Nice value	-3	2

The minimum virtual runtime of the red-black tree is 1,283,642 nanoseconds, the period is set at 20 milliseconds and the scheduler needs to decide which process to execute. Using the guide provided in Appendix A, answer the following questions:

(i) In what order will Process A and B execute within a period? (2 marks)

(ii) What are the time slices allocation for Process A and B within a period? (4 marks)

(iii) What nice value shall you set if you were to make Process B to be allocated more processor time? (2 marks)

(b) Assume that your main memory has a total of 6 frames. Frames #0 to #2 are assigned to Process A and frames #3 to #5 are assigned to Process B. Figure 1 shows the current state of the main memory where frame #5 is empty.

frame #0	A | 2
frame #1	A | 0
frame #2	A | 1
frame #3	B | 2
frame #4	B | 4
frame #5	

Figure 1: Current state of the main memory

Then, the following page trace occurs (process name | page number):

A|3, A|4, A|2, B|6, B|7, B|6, B|2, A|1, A|0, B|4, B|2, B|7

The **Fixed Allocation** and **Local Replacement Scope** are used. Show the steps of each page replacement performed if:

(i) Optimal policy is used. Calculate the total number of page faults for both processes. (8 marks)

Q2. (b)(Continued)

(ii) Clock replacement policy is used. Assume that frames #0 to #4 have use-bit set to 1 and frame # 5 has use-bit set to 0. The clock arrows for Process A and B are pointing at frame #0 and frame #5, respectively. Indicate the use-bit and calculate the total number of page faults for both processes. (9 marks)

[Total : 25 marks]

Q3. (a) Suppose that the disk head has just completed a request at track 125 and is now positioned serving a request at track 100. Then, six requests arrive in the following sequence:

110, 88, 200, 25, 70, 168

(i) Show all the tracks traversed starting from the track 100 based on the SCAN scheduling algorithm and compute the total number of tracks traversed. (4 marks)

(ii) A disk scheduling algorithm suggests that request which is the nearest to the last serviced request will be serviced. What is the name of this suggested policy? (2 marks)

(iii) Calculate the total number of tracks traversed starting from the track 100 using the disk scheduling algorithm that you have named earlier in Q3 (a) (ii). Use the same sequence of track requests given in Q3 (a). (4 marks)

(iv) Is the scheduling algorithm you named in Q3 (a)(iii) suitable for solid-state drive (SSD)? Briefly explain why. (3 marks)

(b) Assume that a secondary storage has a capacity of 1024GB, and the file system block size is 512 Bytes. What is the size (in MB) of the block bitmap if Bit Table disk allocation method is used to manage the free space? Show the steps of your calculation. (4 marks)

(c) Figure 2 shows File B allocated in a disk where the index block is stored in block number 10.

Q3. (c) (Continued)

File Allocation Table

File Name	Index Block
. . .	. . .
File B	10
. . .	. . .

Index Block

Start Block	Length

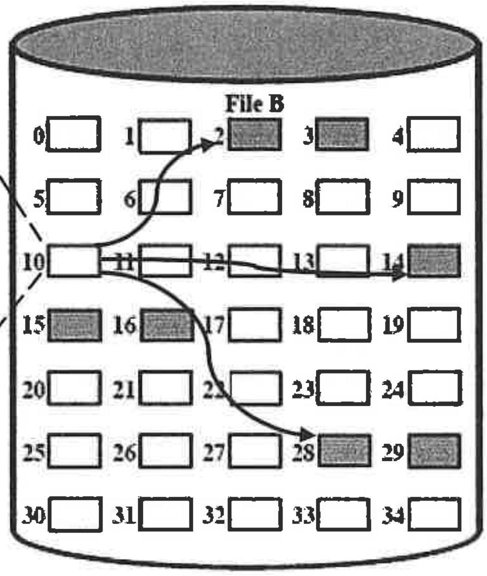

Figure 2: File B blocks allocation in a disk where block number 10 is its index block

(i) According to Figure 2, identify the type of file allocation method used.
(2 marks)

(ii) Fill up the Start Block and Length of the index block shown in Figure 2.
(3 marks)

(iii) Does file consolidation consume lesser Index Block space for the file allocation method shown in Figure 2? Briefly provide your reason.
(3 marks)
[Total : 25 marks]

SECTION B (Choose Any Question)

Q4. (a) Page table will be too large to store in the main memory with given a user space of 8GB, page frame size of 4KB and each page table entry of 8 Bytes.

(i) What is the paging scheme that can solve this problem? (2 marks)

(ii) What are the amounts of memory required to store the page table, with and without the scheme in part Q4 (a)(i)? Show all your calculations. (6 marks)

(b) Given 256KB of memory, show the memory partitioning and allocation in a buddy system for the requests in the following order:

100K, 30K, 1K, 3K, 2K, 20K, 5K (5 marks)

(c) Consider the pseudocode segments of process P and Q in Figure 3a and Figure 3b, respectively. Semaphore *s* and *t* are initialized to 1. Variable *k*, *m*, and *c* are global variables which are shared among process P and Q. Assume that both processes are assigned with the same priority level and there is no other process in the Ready queue.

```
     P(){
P1|     semWait(s);
P2|     m--;
P3|     semWait(t);
P4|     k = m - c;
P5|     semSignal(t);
P6|     semSignal(s);
     }
```

Figure 3a: Process P pseudocode

```
     Q(){
Q1|     semWait(t);
Q2|     m++;
Q3|     semWait(s);
Q4|     k = m + c;
Q5|     semSignal(s);
Q6|     semSignal(t);
     }
```

Figure 3b: Process Q pseudocode

Is it possible for process P and Q to enter deadlock state if:

(i) Process P starts first. (4 marks)

(ii) Process Q starts first. (4 marks)

Note: use the line numbers in Figure 3a and 3b as references in your explanation.

(b) Describe **TWO (2)** disadvantages of using machine instructions to achieve mutual exclusion. (4 marks)

[Total : 25 marks]

Q5. (a) This question is about simple paging using 16-bit logical address. Table 2 shows the first six entries of Process A page table.

Table 2: Process A page table

Page #	Frame #
0	8
1	-
2	5
3	1
4	2
5	7

Given the page size is 2048 Bytes. Answer the following questions:

(i) How many bits of offset field are needed? (2 marks)

(ii) What is the maximum size a process can have in terms of Kilobytes? (2 marks)

(iii) What are the physical addresses corresponding to the following decimal virtual addresses: 8388, 5021, and 2381? Show the translation mechanism and give your final answer in decimal addresses. (6 marks)

(iv) State two advantages if the page size of the system is decreased. (4 marks)

(b) The state of a system at a particular time is shown in Table 3a and Table 3b. The available resource of R1, R2, R3 and R4 are 1, 0, 1 and 1, respectively.

Table 3a: Request matrix

Process	R1	R2	R3	R4
P1	1	1	0	0
P2	1	0	1	0
P3	1	0	0	2
P4	1	0	0	0

Table 3b: Current Allocation

R1	R2	R3	R4
0	1	2	0
0	0	0	1
0	1	0	0
0	0	0	0

(i) Apply deadlock detection algorithm to detect if deadlock will happen. Show the resources needed by each process and the resources available in every stage of your calculation. (7 marks)

(ii) Describe TWO (2) possible deadlock recovery strategies that you can apply to deadlocked processes. (4 marks)

[Total : 25 marks]

APPENDIX A:

Linux Completely Fair Scheduler (CFS):

1. Tasks' time slice allocation:

$$f_i(t) = \frac{w_i}{\sum w_{rbtree}} \times p$$

$f_i(t)$ = time slice received by task i at time t,
w_i = weight of task i

$\sum w_{rbtree}$ = Total weight of tasks in red-black tree

p = period

2. Virtual runtime of task i in n-th iteration:

$$v_i(n) = v_i(n-1) + \left(\frac{\Delta e}{w_i}\right) \times w_{nice\,0}$$

$v_i(n - 1)$ = virtual runtime of task i in previous iteration
Δe = elapsed time task i spent utilizing the processor during iteration n
w_i = weight of task i
$w_{nice\,0}$ = weight assigned to nice 0 task = 1024

3. Mapping of nice value to weight:

```
static constintprio_to_weight[40] = {

/* -20 */     88761,     71755,     56483,     46273,     36291,
/* -15 */     29154,     23254,     18705,     14949,     11916,
/* -10 */      9548,      7620,      6100,      4904,      3906,
/*  -5 */      3121,      2501,      1991,      1586,      1277,
/*   0 */      1024,       820,       655,       526,       423,
/*   5 */       335,       272,       215,       172,       137,
/*  10 */       110,        87,        70,        56,        45,
/*  15 */        36,        29,        23,        18,        15,
}
```